1. Title: Nutritional Profiling and Nutri-Score Prediction of Global Food Products Using Spark SQL and MLlib on a Multi-Node Hadoop Cluster

Course: Big Data Analytics (DSE 3125)

Semester: V

Department: Data Science and Engineering

Section: DSE C

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2. Introduction and Motivation

There is an enormous amount of information about food products available; however, the crowd-sourced datasets contain gaps in nutrients' values, are highly sparse and contain inconsistent data types including different units and text categorizations for nutrients. Processing and modelling millions of such records can be difficult to perform on one computer. While Hadoop HDFS provides a fault-tolerant distributed file system and YARN manages cluster resources, Apache Spark provides an in-memory distributed SQL engine and analytics including machine learning on top of them. This project uses an end-to-end pipeline to build insights from raw food data.

3. Problem Statement

In this project, an analysis of the Open Food Facts product database will be conducted in order to find how nutrient composition (energy, fat, saturated fat, sugars, salt, fiber, proteins) varies according to categories of food and location and to estimate a Nutri-Score of a certain product based on its nutritional properties. In addition, products will be clustered based on their nutrient compositions. All processing will be done on a multi-node Hadoop cluster using Spark on YARN.

4. Objectives

1. Set up a distributed Hadoop cluster (1 NameNode/Resource Manager, 2 DataNodes/Node Managers) and store the dataset in HDFS.
2. Clean and preprocess data using Spark DataFrames: remove duplicate product codes, drop physically impossible nutrient values (less than zero or more than 100g per 100g) and deal with missing data, reporting null counts and removed rows before and after cleaning.

3. Analyse nutritional trends by category and country, and use Spark SQL to identify the top 10 product categories with the highest average sugars, salt and saturated fat.
4. Train and evaluate Spark MLlib models: a Nutri-Score classifier targeting accuracy and weighted F1-score above 0.80, with feature-importance analysis to show which nutrients drive the grade, and a k-means clustering of product nutrient profiles (evaluated by silhouette score).
5. Perform all operations using YARN, recording them in the HDFS, YARN, and Spark interfaces.

5. Dataset Details

Name: Open Food Facts products database. Source: https://world.openfoodfacts.org/data. Format: tab-separated CSV (gzip). Size: approximately 1.2 GB (gzip) and several GB uncompressed; around 4.5 million records (of which approximately 1.4 million have Nutri-Score grade); and 211 attributes per product. License: Open Database License (ODbL) for the database and Database Contents License (DbCL) for the contents; the database does not contain any personal information. Reasons why it is considered big data: The volume is much higher than the required 500 MB, and the database schema is wide and sparse.

6. References

Open Food Facts, “Open Food Facts Data,” https://world.openfoodfacts.org/data.